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 SpaceX Starship Program »
 Starship specs - weight, volumes, etc

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Author**Topic: Starship specs**
☐ **Aeneas**

Full Member



Posts: 203

Germany

Liked: 79

Likes Given: 110

**Re: Starship specs - weight,
 volumes, etc**

 « **Reply #180 on:** 02/26/2021 06:52 am
 »
Quote from:
**TheRadicalModerate on
 04/18/2020 10:41 pm**
**Quote from: dulises on
 04/18/2020 03:16 pm**

Hi everybody, I want to
 go back to the LOX tank
 for landing.

Rafael in the first
 comment arrived to these
 numbers
 LOX spherical header
 tank on nose: 14.56m³
 LOX header tank: 17.9t

Having in mind this
 tweet,
 No, as the header
 tanks are quite small
 relative to main (~30
 tons vs 1200 tons).
 Only uses very tip of
 the nose.

— Elon Musk
 (@elonmusk) April 18,
 2020
[https://twitter.com/
 elonmusk/](https://twitter.com/elonmusk/)

[status/1251386583846023171](#)

it seems that we can assume that the 1200 tons don't include landing fuel. I previously thought 1200 tons were the total fuel.

Also, 30 tons of LOX means at least 27 m³ for the header LOX tank. For LCH₄ it would be about 8 tons of fuel and 18 m³ tank.

I assume using the whole tip of the nose could give us those 30 m³, and there is the main difference, isn't it?

What do you think?

I can't make any sense out of Elon's tweet, either geometrically or from a delta-v standpoint.

If you look at the [Starship PUG](#) that was issued recently, a bit of extrapolation of the diagram of the static payload envelope shows that there's room for at most a 2.5m spherical tank in the nose, which would be 8.18m³.

Now, there are pictures of a LCH₄ spherical header tank located inside the main LCH₄ tank, so we can assume that only the LOX header tank is in the nose.

But that doesn't gibe with a total of 30t of prop in the header tanks. At boiling LOX density (1141 kg/m³), 8.18m³ holds only 9.3t of LOX. At O/F mixture of 3.7, that would be a total header prop load of 11.8t. Last I checked, 11.8 << 30.

Another way to come at this: Figure out the terminal velocity on Mars and use that as a proxy for how much delta-v has to be in the header tanks for EDL. Parameters:

Mars $g=3.71 \text{ m/s}^2$
Landing mass of 120t dry +

150t payload + 30t prop =
300t
470m² aspect in skydiver mode
(just doing 9m x 50m and
eyeballing the fins)
SWAG drag coefficient of 0.45
Mars surface air density of
0.02kg/m³

So terminal velocity should be
 $\sqrt{2mg/(\rho AC_d)} = 725 \text{ m/s}$.

Let's add 200 m/s for attitude
control and that swoopy
skydiver-to-tail-down
maneuver for a total of 925 m/
s. Use Isp=356 for the Starship
center engines (which are
earth sea-level optimized) and
a 120t dry mass + 150t
payload, and we get 82t of prop
needed. 82 >> 30.

Maybe 30t is just for earth EDL?

Earth g = 9.8m/s²
Landing mass = 120t dry + 50t
(max down mass) + 30t prop =
200t
A is still 470m²
Cd=0.45
Air density=1.2kg/m³

Terminal velocity = 124 m/s.
Add 200 m/s for attitude
control, so 324 m/s. Isp=330,
so we need 18t of prop. Not
only is this less than 30t but it's
more than the 11.9t we
calculated geometrically.

Maybe I'm over-budgeting for
attitude control? I can get 217
m/s of delta-v for 9.3t of LOX,
which leaves 93 m/s for
attitude control and the
swoop. But that still doesn't
jibe with the 30t Elon number.

The only thing that seems to
make sense is that there's a
9.3t LOX header tank in the
nose solely for balance, and
the main LCH4 and LOX tanks
both have header tanks in
them as well. Maybe there's a
second LOX header tank
attached to one of the LOX
domes/bulkheads that we
haven't seen? Another
possibility is that there's
always 9.3t of LOX in the nose,
but Mars-bound Starships have

bigger in-main header tanks
than Earth- and lunar-serving
ones.

Any thoughts? Lots of things
not adding up.

Area is due to fins a bit larger
and the drag coefficient must
be much higher 0.7 to 1.0 or
something. A falling barrel is
already at 0.5. Now, you have
spread flaps. So terminal
velocity, I would assume <100
m/s. SN8 indicated 80 m/s.
Landing a 2 g means 160 m/s
for the actual landing burn.
Maybe 3 seconds of hovering
= 30 m/s. belly flop might
need another 50 m/s. Near
ground, there's the need for
240 m/s. Additionally, you
need about 100 m/s for
deorbit. Being conservative
and estimating this all on SL
Isp ~ 20 tons of fuel.

 Logged

 sebk likes this

 **steveleach**

Senior Member



Posts: 3009

Liked: 3523

Likes Given: 1173

**Re: Starship specs - weight,
volumes, etc**

« **Reply #181 on:** 02/26/2021 09:22 am
»

**Quote from: Aeneas on
02/26/2021 06:28 am**

**Quote from: steveleach on
02/25/2021 10:03 pm**

Another thread has
recently got onto the
topic of the mass of
Starship, so I thought it
would be worth bumping
this thread and also
including another
summary. Most of the
numbers are either taken
from Fael's original post
or are pure guesses.

If anyone has a better
number for any of these
elements I'd be more
than happy to update

them.

Item	Tons	Source
===		===
	===	
19 Rings	30.8	Fael
Engine ring	1.3	Fael
Nosecone	9.0	Fael
Bulkheads	9.9	Fael
Front flaps	5.0	Guess
Rear flaps	10.0	Guess
Header	1.3	Fael
tanks		
Downcomer	1.0	Guess
Thrust puck	2.0	Guess
3 x SL	4.5	Fael
Raptors		
Batteries	0.6	beels
Flap	1.0	Guess
actuators		
RCS	0.5	Guess
Legs	1.0	Guess
Stringers	1.0	Guess
Hatches	1.0	Guess
Wiring	0.5	Guess
Avionics	0.5	Guess
COPVs	3.0	Guess
Lifting	1.0	Guess
points		
---	---	---
SN10	80.3	Total
---	---	---
3 x Vac	6.3	Fael
Raptors		
TPS tiles	10.0	Fael
Chomper	5.0	Guess
Payload	5.0	Guess
mount		
Solar	5.0	Guess
panels		
Radiators	5.0	Guess
Full legs	6.0	Guess (extra)
Full RCS	2.0	Guess (extra)
---	---	---
Cargo	124.6	Total
Starship		

THANK YOU! I was looking for this thread quite a bit! I wanted to validate my model. And one of my big questions is, still, the actual flap mass. How comfortable are you with your guess of 10 and 5 for aft and front flaps?

I spent *at least* 3 seconds on those guesses.

Honestly, they aren't based on anything at all, and I'm really hoping that someone with actual aerospace engineering experience can give me a better value.

I believe that SN10 has a mass well over 100 tons though, so we've got to find at least another 20 somewhere.

 Logged

 sebk and Aeneas like this

 **Aeneas**

Full Member



Posts: 203

Germany

Liked: 79

Likes Given: 110

Re: Starship specs - weight, volumes, etc

« **Reply #182 on:** 02/26/2021 09:41 am »

Quote from: steveleach on

02/26/2021 09:22 am

I spent *at least* 3 seconds on those guesses.

Honestly, they aren't based on anything at all, and I'm really hoping that someone with actual aerospace engineering experience can give me a better value.

I believe that SN10 has a mass well over 100 tons though, so we've got to find at least another 20 somewhere.

3 seconds will do...^^

Maybe welds and stringers are more. But I don't see the need for 100+ tons for SN8 and SN9. Terminal velocity was at ~80 m/s. Given that the flaps weren't stretched fully and a drag coefficient of likely below 1, this could be a total mass of 120 tons. And that includes generous landing prop. The explosion of SN9 showed a huge amount of LOX flying everywhere. So 40 tons propellant before bellyflop

burn isn't unrealistic.

 Logged

☐ **Aeneas**

Full Member



Posts: 203

Germany

Liked: 79

Likes Given: 110

Re: Starship specs - weight, volumes, etc

« **Reply #183 on:** 02/26/2021 09:59 am

»

For the optimised Starship, I currently see this:

upper bulkhead	2.1
LCH4 cylinder	3.7
shared bulkhead	2.6
LOX cylinder	8.5
lower bulkhead	3.3
stringers	6.4
upper/lower skirt	5.4
thrust puck	2.2
engines	9.1
flaps+heat shield,	17.4
plumbing, wiring etc?	
heat shield	6.6
(tankage&engines)	
payload bay+heat shield?	13.0
total	80.2

Though I start to think I might be a bit optimistic in regard to flaps and miscellaneous.

Probably the payload connector will add a few tons, too. Nonetheless, in the long run, I would consider <90 tons dry to be realistic. At least for cargo ships. Maybe there will be a bit higher security factor for crewed vehicles.

Steel is 2.8 mm for 1,300 N/mm tensile strength. Max load is 35 m/s. Security factor is 1.6. both tanks have 4.9 bar. The tankage includes 4% ullage and 30 mm of insulation.

 Logged

☐ **steveleach**

Senior Member



Re: Starship specs - weight, volumes, etc

« **Reply #184 on:** 02/26/2021 10:18 am

»

Posts: 3009
Liked: 3523
Likes Given: 1173

OK, so lets try some numbers
on the aft flaps...

I'm using Fael's marked-up
diagram, plus a photo from
Mary of the end of a flap
(attached).

The Flaps are 4.2m wide and
13.87m long.

In the flap-end diagram the
green line is basically the
4.2m, which makes the red
line is 0.8m and the blue line
0.3m (approx).

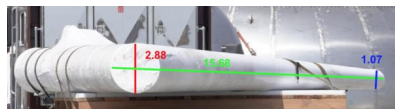
So we have a cylinder 0.8m
diameter and 13.87m long
which is 27.9m area (ignoring
the ends). Assuming 4mm
thick steel with a mass of 8g/
 m^3 that gives us a mass of
0.9 tons.

We then have the top &
bottom skin; they are 4.2m by
13.87m and I've assumed 40%
cut out by the diagonal, so
about $35m^2$. 4mm steel
makes that 1.1 tons, so 2.2
tons for top + bottom.

You then have the bottom side
which is 4.2m by an average of
0.55m deep = $2.31 m^2$, so
0.1 ton. Then the other sides
which are basically a bit over
the flap length, so call it 15m
by 0.3m = $4.5m^2$ = about
0.15 tons.

So that's 0.9 (cylinder) plus 2.2
(top/bottom skin) plus 0.1 plus
0.15 (side skin) = 3.45 tons for
the skin of one flap.

Allow 1.5 tons (wild guess) for
internal structure and you
have 5 tons for an aft flap, so
10 tons for two.



FlapMeasurement.png (523.1 kB, 1328x352 - viewed 352 times.)

Logged

Aeneas likes this

☐ **xvel**

Full Member



Posts: 883

I'm metric and I'm proud of it

Liked: 948

Likes Given: 318

Re: Starship specs - weight, volumes, etc

« **Reply #185 on:** 02/26/2021 04:14 pm

»

Flap skin is not 4mm thick, I'm pretty sure it's below 2mm probably 1mm, they have plenty of 4mm steel in rolls and flat sheets on site and I didn't see it wrinkle like flap skin, in fact it doesn't wrinkle at all. I think the weight of the flaps comes mainly from the internal structure, which we have no idea how it is designed.

Logged

Aeneas likes this

And God said: "Let there be a metric system". And there was the metric system.
And God saw that it was a good system.

☐ **sebk**

Full Member



Posts: 772

Europe

Liked: 974

Likes Given: 27160

Re: Starship specs - weight, volumes, etc

« **Reply #186 on:** 02/26/2021 04:47 pm

»

A little note. There's L2 info on Sn8/Sn9 mass (I'd assume Sn10 is pretty similar). And that info doesn't exactly match what we have here.

So we need quite a bit of grain of salt for the local estimates. Quite likely we're missing a lot of stuff not directly visible.

Logged

☐ **steveleach**

Senior Member



Posts: 3009

Liked: 3523

Likes Given: 1173

Re: Starship specs - weight,**Quote from: sebk on 02/26/2021**« **Reply #187 on:** 02/26/2021 04:49 pm
04:47 pm »

A little note. There's L2 info on Sn8/Sn9 mass (I'd assume Sn10 is pretty similar). And that info doesn't exactly match what we have here.

So we need quite a bit of grain of salt for the local estimates. Quite likely we're missing a lot of stuff not directly visible.

I know, but was trying not to say so explicitly.

☐ **Sciencefan**

Full Member



Posts: 433

Liked: 325

Likes Given: 212

Re: Starship specs - weight, volumes, etc« **Reply #188 on:** 02/26/2021 07:39 pm »**Quote from: sebk on 02/26/2021****04:47 pm**

Quite likely we're missing a lot of stuff not directly visible.

Not after two RUDs. 😊

☐ **Aeneas**

Full Member



Posts: 203

Germany

Liked: 79

Likes Given: 110

Re: Starship specs - weight, volumes, etc« **Reply #189 on:** 02/27/2021 10:56 pm »**Quote from: sebk on 02/26/2021****04:47 pm**

A little note. There's L2 info on Sn8/Sn9 mass (I'd assume Sn10 is pretty similar). And that info doesn't exactly match what we have here.

So we need quite a bit of grain of salt for the local estimates. Quite likely we're missing a lot of stuff not directly visible.

What is L2?

☒ **Coastal Ron****Re: Starship specs - weight,**

Senior Member



Posts: 9933

I live... along the coast

Liked: 11533

Likes Given: 13189

volumes, etc« **Reply #190 on:** 02/27/2021 11:33 pm

»

**Quote from: Aeneas on
02/27/2021 10:56 pm**

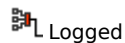
**Quote from: sebk on
02/26/2021 04:47 pm**

A little note. There's L2 info on Sn8/Sn9 mass (I'd assume Sn10 is pretty similar). And that info doesn't exactly match what we have here.

So we need quite a bit of grain of salt for the local estimates. Quite likely we're missing a lot of stuff not directly visible.

What is L2?

Scroll up to the top of this page on NSF and click on the "L2 Sign Up" link. That will take you to an explanation of the paid section of NSF called "L2".



If we don't continuously lower the cost to access space, how are we ever going to afford to expand humanity out into space?

☐ **FPLmanager**

Member

Posts: 1

Liked: 1

Likes Given: 1

Re: Starship specs - weight, volumes, etc« **Reply #191 on:** 09/20/2021 06:20 am

»

Quote

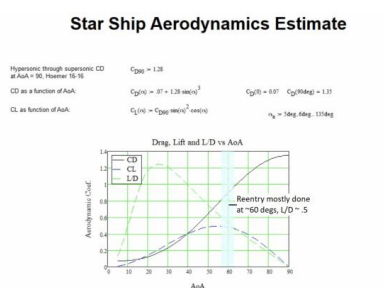
Max L/D, for Mach > 2.5, would be in the neighborhood of 1.2 - 1.3, but Starship was not designed to fly there. It doesn't have the right controls. I believe there will be a minimum aerodynamically controllable AoA (40?). For reentry you really don't want an L/D much higher than about .5.

John

Hi John, and thank you posting the drag/lift estimates from Hoerner's book. I also have a copy, but I can't find the graph you're referring to here in the book. What chapter is it in?

I'm wondering if the equations that you presented are for the whole vehicle (wings+fuselage) or for the fuselage only? I see the similarity to hypersonic flat plate equations (<https://core.ac.uk/download/pdf/42839316.pdf>), so I tend to think they would cover the whole vehicle, but wanted to confirm.

If this is the case, I also wanted to check with you that the correct reference area would still be the projected (=planform) area of the whole vehicle. I believe I have a good estimate on the drag based on your previous post, but I'm not so sure about the lift part, hence making sure I'll use these graphs correctly.



[SSLD.png](#) (1795.56 kB, 1898x1372 - viewed 350 times.)

Logged

TheRadicalModerate likes this

☐ livingjw
Senior Member

Re: Starship specs - weight, volumes, etc
« **Reply #192 on:** 09/21/2021 12:59 am »



Posts: 2382
New World
Liked: 5911
Likes Given: 2928

**Quote from: FPLmanager on
09/20/2021 06:20 am**

Quote

Max L/D, for Mach > 2.5,
would be in the
neighborhood of 1.2 - 1.3,
but Starship was not
designed to fly there. It
doesn't have the right
controls. I believe there
will be a minimum
aerodynamically
controllable AoA (40?).
For reentry you really
don't want an L/D much
higher than about .5.

John

Hi John, and thank you posting
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I'm wondering if the equations
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fuselage only? I see the
similarity to hypersonic flat
plate equations ([https://
core.ac.uk/download/
pdf/42839316.pdf](https://core.ac.uk/download/pdf/42839316.pdf)), so I tend to
think they would cover the
whole vehicle, but wanted to
confirm.

If this is the case, I also wanted
to check with you that the
correct reference area would
still be the projected
(=planform) area of the whole
vehicle. I believe I have a good
estimate on the drag based on
your previous post, but I'm not
so sure about the lift part,
hence making sure I'll use
these graphs correctly.

- I treated the StarShip as a
cylinder at an angle of attack.

- Cd and Cl are based on
length*diameter. For StarShip I
estimate this to be ~ 5000

ft^2 ($\sim 464 \text{ m}^2$). This includes the fins.

- Page 16-16 shows a graph of C_{d0} for a cylinder as a function of Mach.

- Page 18-16 contains the equations for cylinder's C_d and C_l as a function of angle of attack ($\text{Mach} > 2.5$)

- I've added a small C_{d0} for skin friction and base drag.

John

« Last Edit: 09/21/2021 01:01 am by livingjw »

 Logged

 sebk and FPLmanager like this

☐ MGoDuPage

Member

Posts: 66

Liked: 54

Likes Given: 28

Re: Starship specs - weight, volumes, etc

« **Reply #193 on:** 09/09/2022 05:36 pm »

Apologies for the necro-posting, but I searched the forum & didn't see any more recent threads specifically dedicated to this topic.

It's been about a year since the most recent post on this thread, so my WAG is the specifications here might be overdue for an update. (Either from actual design changes and/or previous estimates for certain components/subsystems can be refined due to solid info that's come out in public since September 2021.) If someone has a single reliable & comprehensive link that shows recent "number crunching" (as opposed to just taking vague Elon tweets or glossy SpaceX PR sizzle reels as gospel), then great. But I'm betting it's more likely any new

info is disbursed piecemeal & not necessarily vetted. If so, then forum members here are going to be much better equipped to crowdsource & do that reasonably well vs. the Average Joe (including yours truly).

 Logged

 sebk likes this

☐ **geza**

Full Member



Posts: 754
Budapest



Liked: 492
Likes Given: 90

Re: Starship specs - weight, volumes, etc

« **Reply #194 on:** 09/12/2022 04:35 pm »

It is a good idea to keep this thread uo-to-day.

I have question. It is usually assumed that all rings have an equal height, 6 foot, equal to the height of the coil of stainless. However, the rings on the photos of the real thing does not look like, as of equal height.

Can anybody clarify?

 Logged

☐ **Nomadd**

Senior Member



Posts: 9104
Virginia

Liked: 61610
Likes Given: 1411

Re: Starship specs - weight, volumes, etc

« **Reply #195 on:** 09/12/2022 05:43 pm »

Quote from: geza on 09/12/2022 04:35 pm

It is a good idea to keep this thread uo-to-day.

I have question. It is usually assumed that all rings have an equal height, 6 foot, equal to the height of the coil of stainless. However, the rings on the photos of the real thing does not look like, as of equal height.

Can anybody clarify?

Not all the circumferential lines you see are ring joins. There are structural hoops welded inside and places where the tank domes meet the walls.

« Last Edit: 09/12/2022 05:44 pm by Nomadd »

 Logged

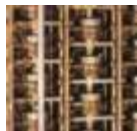
 sebk likes this

Those who danced were thought to be quite insane by those who couldn't hear the music.



FutureSpaceTourist

Global Moderator
Senior Member



Posts: 57751
UK



Liked: 94848
Likes Given: 44765

Re: Starship specs - weight, volumes, etc

« **Reply #196 on:** 05/24/2023 06:44 pm »

twitter.com/elonmusk/
status/1661441658473570304

Quote

Starship payload is 250 to 300 tons to orbit in expendable mode.

Improved thrust & lsp from Raptor will enable ~6000 ton liftoff mass.

Assuming Starship still has a liftoff TWR of 1.5 and 33 Raptor engines, that assumes 272 tf per engine which I guess it is only enabled by using the upcoming Raptor 3 or whatever name they put to it. <https://t.co/2euExf1goR>
— Alejandro Alcantarilla Romera (Alex) (@Alexphysics13) May 24, 2023
<https://twitter.com/alexphysics13/status/1661442792005025792>

Quote

Assuming Starship still has a
liftoff TWR of 1.5 and 33 Raptor
engines, that assumes 272 tf
per engine which I guess it is
only enabled by using the
upcoming Raptor 3 or whatever
name they put to it.

 Logged

 CruddyCuber, Robotbeat,
chief and 2 others like this


FutureSpaceTourist

Global Moderator
Senior Member



Posts: 57751
UK



Liked: 94848
Likes Given: 44765

Re: Starship specs - weight, volumes, etc

« **Reply #197 on:** 07/10/2023 06:30 am »

twitter.com/elonmusk/
status/1678276840740343808

Quote

Looks like we can increase
Raptor thrust by ~20% to reach
9000 tons (20 million lbs) of
force at sea level

And deliver over 200 tons of
payload to a useful orbit with
full & rapid reusability.

50 rockets flying every 3 days
on average enables over a
megaton of payload to orbit
per year – enough to build a
self-sustaining city on Mars.

— Elon Musk (@elonmusk) July
10, 2023

[https://twitter.com/elonmusk/
status/1678278811186544640](https://twitter.com/elonmusk/status/1678278811186544640)

Quote

And deliver over 200 tons of
payload to a useful orbit with
full & rapid reusability.

50 rockets flying every 3 days
on average enables over a
megaton of payload to orbit per
year – enough to build a self-
sustaining city on Mars.

 Logged sebk, M.E.T., RedLineTrain
and 1 other like this☐ **M.E.T.**

Senior Member



Posts: 2582

Liked: 3137

Likes Given: 564

Re: Starship specs - weight, volumes, etc« **Reply #198 on:** 07/10/2023 07:40 am

»

So is it more beneficial to increase overall thrust by 20%, or to keep thrust the same but reduce the number of engines required by 20%?

In the latter case I presume the payload gain is more or less equivalent to the weight saved by removing the excess engines, while in the former case the rocket needs to be stretched to accommodate more fuel, correct?

 Logged☐ **volker2020**

Full Member



Posts: 356

Frankfurt, Germany

Liked: 378

Likes Given: 961

Re: Starship specs - weight, volumes, etc« **Reply #199 on:** 07/10/2023 09:35 am

»

**Quote from: M.E.T. on
07/10/2023 07:40 am**

So is it more beneficial to increase overall thrust by 20%, or to keep thrust the same but reduce the number of engines required by 20%?

In the latter case I presume the payload gain is more or less equivalent to the weight saved by removing the excess engines, while in the former case the rocket needs to be stretched to accommodate more fuel, correct?

Yes. The faster the rocket get's to speed, the lower the gravity losses are. And they eat up a lot of your total delta-v.

 Logged



InterestedEngineer likes this

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